

Packaging the information

Inside the nucleus of every cell in the human body are 46 molecules of DNA, carrying all the information needed to build and maintain a human being. Each molecule is shaped like a twisted ladder—named a double helix—and packaged up into a bundle called a chromosome. Genetic information is carried by the sequence of different chemical units, known as bases, that make up the “rungs” of the ladder.

Chromosome

A tightly packed mixture of protein and DNA forms a chromosome. Each chromosome contains one long molecule of DNA. The DNA in this chromosome has replicated to make an X-shape. It is ready to split in two, sending one molecule to one cell and one molecule to another.

Cell nucleus

The nucleus of each cell is where genetic information is stored.

Protein

The DNA double helix is wrapped around balls of protein to help it fit inside the cell.

Building blocks

The sides of all DNA molecules are always the same—made up of alternating chemical blocks of sugar (large balls) and phosphate (small balls).

Genetics and DNA

The characteristics of a living thing—who we are and what we look like—are determined by a set of instructions carried inside each of the body’s cells.

Instructions for building the body and keeping it working properly are held in a substance called DNA (deoxyribonucleic acid). The arrangement of chemical building blocks in DNA determines whether a living thing grows into an oak tree, a human being, or any other kind of organism. DNA is also copied whenever cells divide, so that all the cells of the body carry a set of these vital genetic instructions. Half of each organism’s DNA is also passed on to the next generation in either male or female sex cells.

GENES ON A SECTION OF DNA



What is a gene?

The information along DNA is arranged in sections called genes. Each gene has a unique sequence of bases. This sequence acts as a code to tell the cell to make a specific protein, which, in turn, affects a characteristic of the body.

